DEEP Metrics

Western Gulf Coastal Plain reference assessment (western-gulf-coastal-plain@1)

Stream	Mink Brook
Coordinates	43.68581234, -72.23671234
COMID / HUC8	9327042 / 01080104
Drainage area	12.43 km2
Reach length	1000.0 ft
Watershed basis	HR reach watershed (STAF site engine v0.5.0)
StreamCat reach	COMID 9327042 (this reach)

Function	Metric	F/D	How it is measured	Status	Value	Source
Catchment hydrology	Impervious surface (%)	D	Percent of the delineated watershed in impervious surface, computed by DEEP. No field measurement.	Available	3.0	NLCD 2021
Catchment hydrology	Agriculture, crop and hay (%)	D	Percent of the delineated watershed in cultivated crops plus hay and pasture, computed by DEEP. No field measurement.	Available	12.75	EPA StreamCat
Surface water storage	Wetland (%)	D	Percent of the watershed in woody plus herbaceous wetland (NLCD 2019), read from EPA StreamCat for the reach. No field measurement.	Reference only	0.4	Computed by DEEP
Reach inflow	Road density (km/sq km)	D	Road length per square kilometer of the delineated watershed, computed by DEEP. No field measurement.	Available	3.0	NLCD 2021
Streamflow regime	Base flow index (%)	D	Percent of streamflow that comes from base flow in the watershed (USGS base flow index grid), read from EPA StreamCat for the reach. No field measurement.	Available	12.75	EPA StreamCat
Streamflow regime	Degree of regulation (%)	D	Normal storage of upstream dams as a percent of mean annual runoff, both read from EPA StreamCat for the reach. No field measurement.	Available	0.4	Computed by DEEP
Low flow and baseflow dynamics	Base flow index (%)	D	Percent of streamflow that comes from base flow in the watershed (USGS base flow index grid), read from EPA StreamCat for the reach. No field measurement.	Available	12.75	EPA StreamCat
Low flow and baseflow dynamics	Residual pool depth (cm)	F	Measure thalweg depth at 10 or 15 evenly spaced stations between each pair of transects (100 or 150 per reach). Value: mean residual pool depth in cm, from that profile and the reach slope (NRSA physical habitat calculations).	Entered	7.5	
High flow dynamics	Impervious surface (%)	D	Percent of the delineated watershed in impervious surface, computed by DEEP. No field measurement.	Available	3.0	NLCD 2021
Floodplain connectivity	Bankfull height above channel (m)	F	At each transect, measure the bankfull height above the present water surface in m, with a rod at the water's edge and a clinometer held level at bankfull. Value: mean of the 11 heights.	*Measure in the field*		
Floodplain connectivity	Bankfull width:depth ratio	F	At each transect, measure bankfull width, bankfull height above the water surface and thalweg depth. Value: mean bankfull width divided by mean bankfull depth (bankfull height plus thalweg depth).	*Measure in the field*		

Function	Metric	F/D	How it is measured	Status	Value	Source
Hyporheic connectivity	Embeddedness (%)	F	At 5 evenly spaced points on each of the 11 transects, estimate the percent of the particle buried in sand or finer sediment. Value: mean of the 55 estimates.	*Not applicable*		
Channel evolution	Bankfull height above channel (m)	F	At each transect, measure the bankfull height above the present water surface in m, with a rod at the water's edge and a clinometer held level at bankfull. Value: mean of the 11 heights.	*Measure in the field*		
Channel evolution	Bankfull width:depth ratio	F	At each transect, measure bankfull width, bankfull height above the water surface and thalweg depth. Value: mean bankfull width divided by mean bankfull depth (bankfull height plus thalweg depth).	*Measure in the field*		
Channel and floodplain dynamics	Sinuosity	F	Take a compass backsight bearing and the distance between each pair of neighboring transects. Value: channel length along those segments divided by the straight-line distance between the reach ends.	*Measure in the field*		
Channel and floodplain dynamics	Bank angle (degrees)	F	On both banks of each transect, lay a pole about 1 m long against the bank and read its angle from horizontal with a clinometer, in degrees. Value: mean of the 22 readings.	Entered	12.5	
Sediment continuity	Relative bed stability (log)	F	Computed: the base 10 log of the geometric mean particle diameter divided by the largest diameter the bankfull flow can move, from the particle count, reach slope, bankfull depth, wood tally and residual pools (NRSA physical habitat calculations).	*Measure in the field*		
Bed composition and bedform dynamics	Embeddedness (%)	F	At 5 evenly spaced points on each of the 11 transects, estimate the percent of the particle buried in sand or finer sediment. Value: mean of the 55 estimates.	*Not applicable*		
Bed composition and bedform dynamics	Large wood volume (m3/100m)	F	Between each pair of transects, tally every piece of wood at least 10 cm in diameter and 1.5 m long in or above the bankfull channel, by length and diameter class. Value: wood volume per 100 m of channel, from the tally.	*Measure in the field*		
Light & thermal regime	Canopy density, mid-channel (%)	F	At mid-channel on each transect, read a convex spherical densiometer facing upstream, downstream, left and right (covered points of 17). Value: mean of the 44 readings, as a percent.	*Measure in the field*		
Carbon processing	Riparian veg cover, woody (frac)	F	In a 10 m by 10 m plot on each bank at each transect, estimate the areal cover class of woody vegetation in the canopy, understory and ground layer. Value: summed woody cover of the 3 layers, averaged over the 22 plots, as a fraction.	*Measure in the field*		
Nutrient cycling	Total nitrogen (mg N/L)	F	Total nitrogen (mg N/L), lab analysis of the unfiltered mid-channel grab sample from the X-site, kept on ice.	Entered	17.5	
Nutrient cycling	Total phosphorus (ug/L)	F	Total phosphorus (ug/L), lab analysis of the mid-channel grab sample from the X-site, kept on ice.	*Measure in the field*		
Water & soil quality	Specific conductivity (uS/cm)	F	Specific conductance (uS/cm). NRSA reports the lab value of a mid-channel grab sample at the X-site (the reach center); a calibrated multi-parameter meter reads the same quantity in the stream there.	*Measure in the field*		

Function	Metric	F/D	How it is measured	Status	Value	Source
Habitat provision	Natural fish cover (frac)	F	In the 10 m of channel centered on each transect, estimate the areal cover class of each fish cover type on the form (wood, brush, overhanging vegetation, undercut banks, boulders, others). Value: summed cover of the natural types, averaged over the transects.	*Measure in the field*		
Population support	Fish MMI (MMI points)	F	Computed: from the upstream electrofishing pass (every habitat between the transects, every fish identified and counted), score EPA's 8 regional fish metrics, several adjusted for watershed area, and rescale to 0 to 100 (NRSA 2018-19 technical support document).	*Measure in the field*		
Community dynamics	Benthic macroinvertebrate MMI (MMI points)	F	Computed: from the 11-transect composite D-frame sample (500 um mesh) and its fixed-count lab subsample, score EPA's 6 regional benthic metrics against the published floors and ceilings and rescale the sum to 0 to 100 (NRSA 2018-19 technical support document).	Entered	22.5	
Watershed connectivity	Mapped dams within one mile (dams)	D	Dams in the USACE National Inventory of Dams within 1 mile of the site, counted by DEEP. Note in the field any barrier the inventory does not show.	Available	12.75	EPA StreamCat
Community dynamics	EPT taxa richness			*Not scored*		Insufficient reference support
Community dynamics	Shannon diversity (H')			*Not scored*		Insufficient reference support
Community dynamics	Tolerant individuals (%) ~ HBI			*Not scored*		Insufficient reference support
Population support	Benthic total taxa richness			*Not scored*		Insufficient reference support
Light and thermal regime	Chlorophyll a (ug/L)			*Not scored*		Insufficient reference support
Water and soil quality	pH (Std. Units)			*Not scored*		Insufficient reference support
Water and soil quality	Turbidity (NTU)			*Not scored*		Insufficient reference support
Population support	Native fish taxa richness			*Not scored*		Insufficient reference support
Bed composition and bedform dynamics	Substrate size (log mm)			*Not scored*		Insufficient reference support
Habitat provision	Fast-water / riffle habitat (%)			*Not scored*		Insufficient reference support
Hyporheic connectivity, Sediment continuity	Sand + fines (%)			*Not scored*		Insufficient reference support

F is measured in the field and D is answered from the desk. Available: the value was pulled and is printed on the field worksheet. Unavailable: no source answered, so measure or estimate it on site. Reference only: the value is shown and kept out of the score, because the curve was fitted on another data source. Not scored: the assessment withholds the metric for insufficient reference support.